DO NOW — RECALL

Lesson 096 · Little Man Computer: Instructions and the A

Name: _____ Class: _____ Date: _____

1. What is the ACC for?

2. What does STA X do?

3. What does INP do?

4. Half adder A1,B1: what Sum and Carry?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Stores current/intermediate calculation.

2. Model answer

Stores current ACC into mailbox X.

3. Model answer

Reads input into ACC, replacing its former value.

4. Model answer

Sum0 (XOR); Carry1 (AND).

By the end, you can…

• LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

• INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

• OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Vocabulary — know these terms

LMC — Little Man Computer, simplified stored-program model.

mailbox — Memory location.

ACC — Accumulator for working results.

STA — Store ACC into memory.

LDA — Load ACC from memory.

NEW LEARNING

Read, annotate and explain

Observe

LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

Apply

INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

Evaluate

OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

Follow the reasoning

FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace program above for inputs 4,7: ACC after each INP, STA and ADD; output. [6]

Question 2

Write an LMC program that reads one integer, stores it as N, then loads and outputs N. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare LDA FIRST, STA FIRST and ADD FIRST in relation to ACC and mailbox; give distinct effects. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

STA FIRST overwrites ACC with the mailbox's value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.